

Root Physics via Latin & Greek

CON-/COM- *together, with* (L. *cum*)

convergence, concentration, convex, concave, conductor, condensation, conservation, coincide, coherent, constructive interference

EQUI- *equal* (Latin *aequus*)

equilibrium, equipartition, equivalent, equidistant, equipotential, equation

THERMO-/THERM- *heat* (G. *therme*)

thermodynamics, thermometer, thermal equilibrium, thermocouple, isotherm, exothermic, endothermic

DYNA-/DYNAM- *power, force* (Greek *dynamis*)

dynamics, dynamo, aerodynamics, hydrodynamics

POLE-prefixes *number + pole/end* (Greek/Latin numerals)

monopole, dipole, bipolar, quadrupole, multipole, bimetallic strip

-METER/-METRY *measure* (Greek *metron*)

thermometer, barometer, manometer, ammeter, voltmeter, calorimetry, symmetry

ISO- *equal, same* (Greek *isos*)

isotope, isotherm, isobar, isochoric, isotropic

-TROPE/-TROPIC *turning* (G. *tropos*)

isotropic, anisotropic

-STATIC/-STASIS *standing still* (Greek *statikos/stasis*)

electrostatic, hydrostatic, homeostasis

RE- *back, again* (Latin)

reflection, refraction, resistance, resonance, coefficient of restitution, repulsion

TRANS- *across* (Latin)

transverse, transmission, transformer, translational motion

-SCOPE/-SCOPY *see* (G. *skopein*)

oscilloscope, spectroscope, telescope

PHOTO- *light* (G. *phos/photos*)

photon, photoelectric, photodiode

HOMO-/HETERO- *same/different* (Greek)

homogeneous, heterogeneous

SUPER-/SUB- *above/below* (Latin)

superposition, superconductor, subsonic

UNI-/MULTI- *one/many* (Latin)

uniform, unipolar, multimeter

-VERS/-VECT *turn* (Latin *vertere*)

vector, inverse, converse, reversible

AMPL- *large* (Latin *amplus*)

amplitude, amplifier

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Test Yourself

1. A lens that curves outward, bulging in the middle, is described as _____. Which root gives you this, and what does the root mean?
2. Two waves that stay in a fixed phase relationship to each other are called _____. Which root family does this word belong to?
3. A magnet with two distinct ends behaves as a _____, while a single isolated magnetic charge (never observed, but predicted by some theories) would be a _____.
4. Explain, using the root, why a material at *thermal equilibrium* with its surroundings has no net heat flow.
5. An *isotropic* radiator emits light equally in all directions. Break the word into its two roots and explain what each part contributes to the meaning.
6. A voltmeter, ammeter, and barometer all share a suffix. What does that suffix mean, and what does each prefix (*volt-*, *am(pere)-*, *baro-*) tell you the instrument measures?
7. Explain the difference between *isotope* and *isotherm* despite both starting with the same root.
8. *Superposition* and *superconductor* both use the prefix *super-*. Does it mean the same thing in both words? Justify your answer with reference to what physically happens in each case.
9. Why might *equipartition of energy* be a useful name for the idea that energy is shared equally between degrees of freedom in an ideal gas? Use the root to justify your answer.
10. Give one word from the sheet above that you had not consciously linked to its root before today, and write your own one-sentence definition based only on the root meaning (no dictionary allowed).

Answer Bank

1. **Convex.** From *con-* (together) combined with a root meaning "arched"; a convex surface curves outward as if gathered up into a dome.
2. **Coherent.** From *con-/co-*, "together": coherent waves keep a constant phase relationship with each other.
3. Two ends: **dipole**. Single isolated charge: **monopole**. *di-* = two, *mono-* = one.
4. Equilibrium comes from *equi-* (equal); at thermal equilibrium the heat flow one way equals the heat flow the other way, so the net flow is zero.
5. *Iso-* (equal/same) + *-tropic* (turning, direction): equal in every direction, so the emission doesn't depend on which way you look.
6. The suffix *-meter* means "measure" (Greek *metron*). Volt-meter measures voltage, am(pere)-meter measures current, baro-meter measures atmospheric pressure (*baros* = weight/pressure).
7. Both start with *iso-* (equal/same), but the second root differs: *-topy* relates to "place" (same place in the periodic table, different neutron number), while *-therm* relates to heat (same temperature).
8. Not quite. In *superposition*, *super-* means "on top of/above," describing waves adding on top of each other. In *superconductor*, *super-* means "beyond/exceeding normal," describing resistance dropping beyond the ordinary limit, effectively to zero.
9. *Equi-* (equal) + *partition* (division into parts): the energy is divided up equally between each available degree of freedom.
10. Open answer — check that the definition follows logically from the root meanings, even if the wording differs from the dictionary.

Matching Quiz: Prefix to Meaning

Draw a line (or write the matching letter) connecting each prefix/root on the left to its correct meaning on the right. Meanings are deliberately shuffled and there are more meanings than prefixes, so some will be left over.

Prefix/Root		Meaning
1. CON-/COM-	_____	A. measure
2. ISO-	_____	B. light
3. TRANS-	_____	C. together, with
4. -METER	_____	D. equal, same
5. PHOTO-	_____	E. standing still
6. RE-	_____	F. across
7. -STATIC/-STASIS	_____	G. back, again
8. DYNA-/DYNAM-	_____	H. above, beyond
9. SUPER-	_____	I. power, force
10. UNI-/MULTI-	_____	J. one, many
		K. turning
		L. see

Bonus: pick one leftover meaning from the right-hand column and name a physics word that uses it, even if the prefix isn't listed on the left.

Tricky Words: Construct Your Own Definition

These words haven't appeared on the sheet yet. For each, you're given the root meanings; use them to build your own definition before checking any textbook or dictionary. Some of these are deliberately awkward, since the roots pull in directions you might not expect.

1. **Adiabatic** (*a-* = not, *dia-* = through, *-batic* from *bainein* = to pass/go)

Your definition:

2. **Diathermic** (*dia-* = through, *therm* = heat)

Your definition:

3. **Antinode** (*anti-* = opposite, *node* from Latin *nodus* = knot)

Your definition:

4. **Centripetal** (*centri-* = centre, *-petal* from *petere* = to seek)

Your definition:

5. **Monochromatic** (*mono-* = one, *chroma* = colour)

Your definition:

6. **Synchrotron** (*syn-* = together, *chronos* = time, *-tron* = instrument/device)

Your definition:

7. **Quasi-static** (*quasi-* = as if, almost, *static* = standing still)

Your definition:

8. **Ambipolar** (*ambi-* = both, *polar* from pole)

Your definition: